

CFD Forge: A Physics-Constrained Agentic Framework for Autonomous CFD

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Abstract

Computational fluid dynamics (CFD) requires substantial manual coordination across geometry preparation, meshing, solver setup, diagnosis, and interpretation. Repeating this work across design alternatives limits high-throughput screening and rapid engineering iteration. We present CFD Forge, a programmable agentic CFD framework built around OpenFOAM. The framework connects engineering requests and geometry parameters to case construction, meshing, boundary and initial conditions, solver execution, diagnostics, visualization, and reporting. A large language model reasons across simulation evidence, interprets flow behavior, identifies likely causes of problems, and proposes bounded corrective actions. Modular scientific contracts complement this reasoning with deterministic measurements and acceptance criteria, retaining final ACCEPT, REJECT, or INCONCLUSIVE authority. Demonstrations include a converging-diverging nozzle and a two-dimensional Mach-2 forward-facing step that satisfy their registered criteria. A three-dimensional turbulent cube serves as a diagnostic stress test: quantitative wake analysis exposes continuing lateral-mode development despite comparatively stable mean quantities, preventing premature acceptance. Case files, diagnostic evidence, action histories, and field-evolution visualizations make the workflow auditable. CFD Forge brings execution, reasoning, and scientific assessment into a common workflow, with future applications in high-throughput simulation, active learning, and engineering design optimization.

1 Introduction

Computational fluid dynamics (CFD) is central to engineering analysis, but obtaining a useful result requires a sequence of decisions and operations. An analyst prepares a CAD model or parameterized geometry, defines the domain, generates a mesh, selects physical models, and assigns boundary and initial conditions. Solver execution is followed by diagnosis, assessment, interpretation, and reporting. Although individual steps can be scripted, coordinating them remains expertise-intensive. The NASA CFD Vision 2030 study identifies workflow automation as an important requirement for routine computational analysis and design [1].

This effort becomes a bottleneck in high-throughput screening and iterative design, where many geometries or operating conditions must be evaluated consistently [3, 5]. A programmable CFD workflow could turn an engineering objective into a sequence of reusable operations, then return both simulation results and an explanation of their significance. Quantities such as drag, lift, or

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pressure loss could subsequently guide new evaluations in an active-learning or optimization loop. Recent LLM-assisted CFD optimization work illustrates this broader direction [4].

Large language models (LLMs) offer a way to coordinate such operations through natural-language interpretation, reasoning, and tool use [6, 7], supported by retrieval of technical information [8]. OpenFOAM provides a programmable solver foundation [2]. MetaOpenFOAM and MetaOpenFOAM 2.0 automate simulation and post-processing [9, 10]; OpenFOAMGPT and OpenFOAMGPT 2.0 develop retrieval-assisted and end-to-end workflows [11, 12]. CFD-specific fine-tuning [13], Foam-Agent [14], and ChatCFD [15] provide further approaches to automation. AI CFD Scientist and PhyNiKCE also address physical reasoning and verification [18, 19]. These efforts motivate closer integration of workflow execution, diagnosis, and scientific assessment.

That integration requires attention to the meaning of a successful simulation. Solver completion, small residuals, or plausible contours can coexist with incorrect boundary conditions, unresolved numerical errors, or a flow that is still developing. Verification and validation therefore require multiple sources of evidence [16, 17]. An agent must interpret these signals together and distinguish a useful next action from evidence sufficient to accept the result.

CFD Forge connects engineering intent and geometry parameters to case construction, meshing, boundary and initial conditions, OpenFOAM execution, diagnostics, corrective actions, and reporting. The LLM reasons across solver logs, field diagnostics, conservation measures, and flow histories to interpret behavior, identify likely causes of problems, and propose what should be tried next. Deterministic scientific checks measure explicit criteria, enforce thresholds, and retain final **ACCEPT**, **REJECT**, or **INCONCLUSIVE** authority. This division gives reasoning a substantive role while keeping acceptance reproducible. Modular scientific contracts organize physical assumptions, numerical requirements, and allowed actions so that additional simulation families can be incorporated. The current evaluation uses registered families and parameterized geometries.

We demonstrate the workflow with a compressible nozzle assessed against quasi-one-dimensional theory and a Mach-2 forward-facing step assessed for transient numerical health and conservation. A three-dimensional turbulent cube provides a diagnostic stress test. Although execution completes and mean drag becomes comparatively stable, quantitative force and wake diagnostics reveal continuing lateral-mode growth. The framework uses this evidence to prevent premature acceptance under the stationarity criterion, illustrating why several physical signals must be examined together. The contribution is a programmable CFD workflow that combines evidence-based reasoning with explicit scientific assessment and preserves its decisions through case files, diagnostic records, action histories, visualizations, and reports. This architecture provides a basis for larger simulation campaigns and future integration with active-learning and design-optimization loops.

2 CFD Forge Framework and Scientific Assessment

CFD Forge organizes CFD analysis as a programmable workflow that connects an engineering request to simulation execution, assessment, and reporting. The framework combines language-model reasoning with programmatic case construction and explicit scientific checks. Its central design principle is to separate proposing an interpretation or action from determining whether the resulting simulation satisfies the prescribed acceptance criteria.

2.1 Workflow Overview

The workflow begins with a natural-language engineering request describing the intended flow problem and relevant geometry or operating parameters. The request is interpreted and matched to a

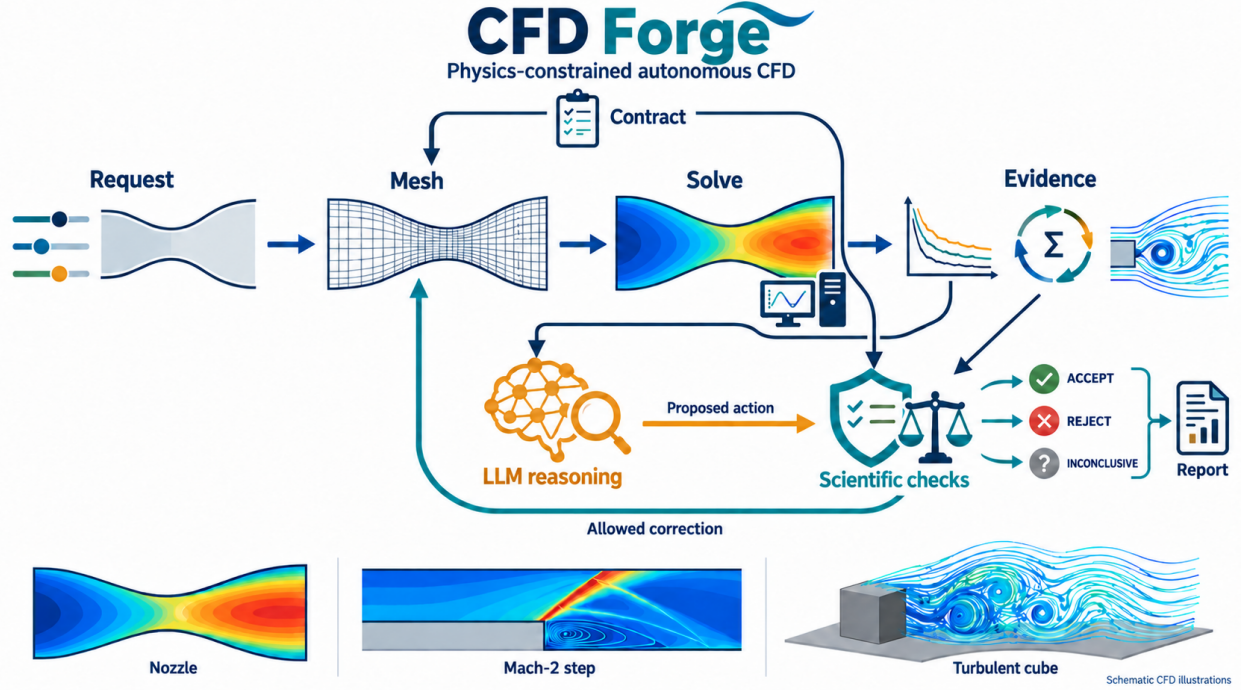


Figure 1: Functional organization of CFD Forge. Language-model outputs are proposals; deterministic checks retain authority over admissibility, permitted actions, and final acceptance. Family-specific runners implement case construction, solver execution, and corrective-action loops.

registered simulation family. Each family supplies a scientific contract defining the physical assumptions, admissible inputs, numerical recipe, required diagnostics, and permitted corrective actions.

For an admissible request, a family-specific runner constructs the case, generates the mesh, assigns boundary and initial conditions, and executes OpenFOAM. Diagnostic routines then extract evidence from the mesh, solver logs, saved fields, and time histories. This evidence supports both model-assisted interpretation and deterministic scientific assessment. Where an allowed corrective action is justified, the runner can update the permitted settings or continue execution before re-assessing the result. Otherwise, the workflow terminates with a recorded decision and supporting artifacts.

The present implementation uses parameterized geometries and registered CFD families. This provides a concrete interface between engineering intent and executable simulation operations, while allowing additional families to be incorporated through their own scientific contracts and runners.

2.2 Reasoning and Scientific Authority

The LLM provides a flexible interface for interpreting engineering requests and reasoning over simulation evidence. Its responsibilities include explaining observed behavior, identifying likely causes of problems, proposing a next action, and supporting the engineering narrative. The evidence supplied to this reasoning process can include solver-health indicators, field extrema, conservation measures, convergence histories, and family-specific diagnostics.

Scientific acceptance is evaluated separately. Deterministic routines calculate the relevant quantities and compare them with the registered criteria. The LLM can propose that a run appears complete or requires further work, but its statement does not override a failed or unresolved check.

Similarly, a proposed correction must be permitted by the family contract before it can be applied. This separation yields three terminal outcomes:

- **ACCEPT**: the required checks are satisfied.
- **REJECT**: the evidence violates a required criterion.
- **INCONCLUSIVE**: the available evidence is insufficient to resolve the required assessment.

Acceptance therefore refers to the stated scientific contract and the evidence available for that run. It does not imply that every source of modeling or numerical uncertainty has been eliminated.

2.3 Scientific Contracts and Corrective Actions

A scientific contract makes the basis for assessment explicit before a result is accepted. It specifies the geometry and operating range, physical model, boundary-condition formulation, mesh requirements, numerical controls, diagnostic quantities, and reference comparisons applicable to a family. The criteria reflect the intended flow regime: a steady nozzle requires appropriate stationarity evidence, whereas an unsteady compressible step requires conservation checks that account for changes in stored mass.

Corrective actions are restricted to those supported by the registered workflow. The model proposes an action, while deterministic checks establish whether it is admissible and whether another solver attempt is warranted. The resulting case is then assessed using the same acceptance criteria rather than a revised threshold chosen to match the outcome.

The archived nozzle demonstration illustrates this sequence: unmet stationarity and mass-balance criteria led to a continuation, followed by reassessment and acceptance. Such records show how evidence informs execution decisions. They do not, by themselves, establish that model-assisted diagnosis outperforms a fixed-rule controller; that comparison requires a separate evaluation.

2.4 Evidence, Reporting, and Reproducibility

CFD Forge retains the artifacts needed to inspect both the simulation and the decisions made around it. These include case configurations, mesh information, solver logs, diagnostic measurements, model-call records where available, proposed actions, and final assessments. Plots, contours, and field-evolution videos connect numerical summaries to the computed flow behavior. Field-evolution visualizations use saved CFD fields at recorded simulation times.

The framework distinguishes live execution from replay of archived evidence. Replay supports inspection and regression checking without claiming that a new simulation or model call occurred. Likewise, deterministic-backend execution is distinguished from execution using an LLM. These distinctions are necessary when interpreting workflow results and attributing an action to model reasoning.

Together, the execution records and scientific assessments provide an auditable account of what was requested, what was computed, which actions were taken, and why the result received its final decision. The following sections examine this workflow through the nozzle, forward-facing-step, and turbulent-cube studies.

3 Experimental Setup and Assessment Protocol

We examine CFD Forge through two compressible-flow demonstrations and a three-dimensional turbulent-flow diagnostic study. The converging-diverging nozzle provides an analytical consistency

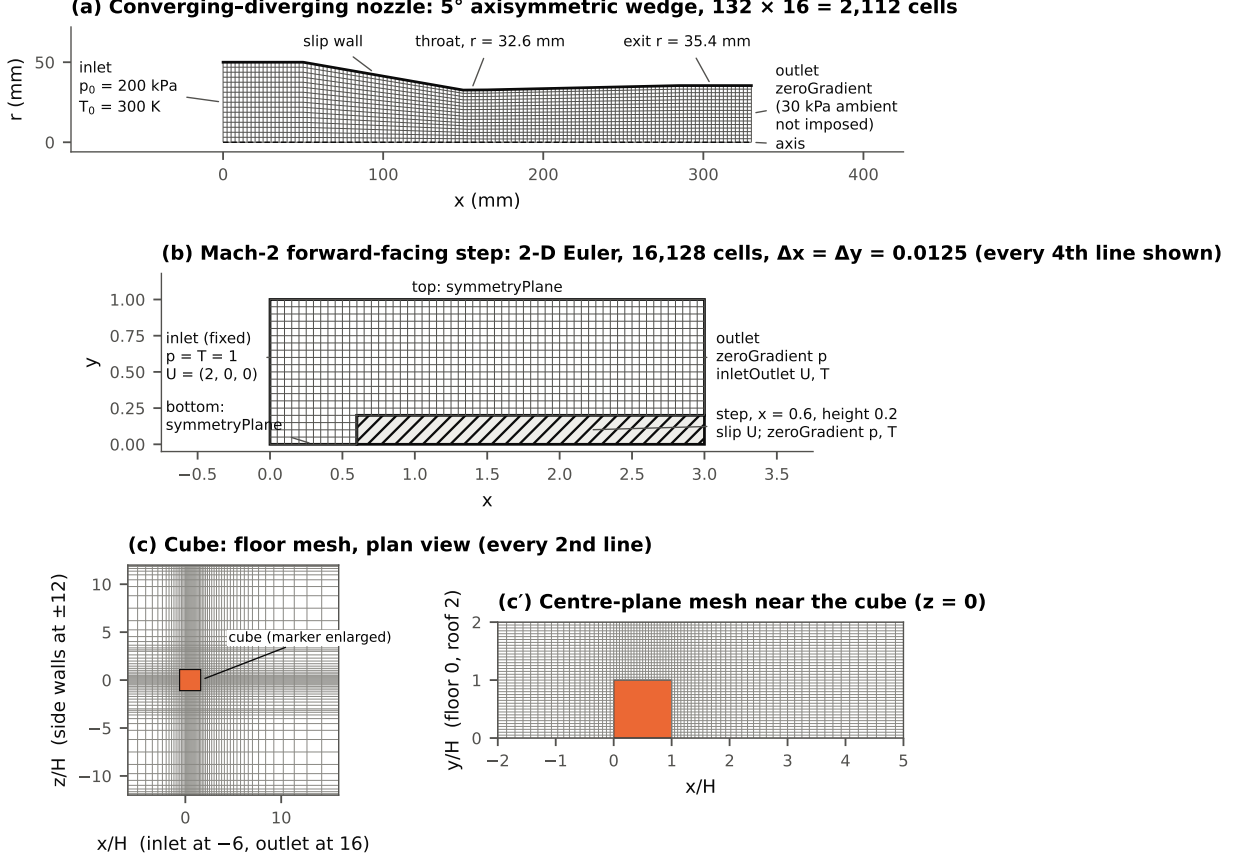


Figure 2: Computational configurations, drawn from the preserved case definitions. (a) Nozzle wedge mesh in the x - r plane, reconstructed exactly from the archived `blockMeshDict` (five linear blocks, uniform grading). (b) Forward-facing-step domain and block structure; every fourth grid line is shown. (c) Floor plan (every second line; cube marker enlarged) and (c') centre-plane views of the cube mesh, plotted from the grid coordinates in the preserved `constant/polyMesh/points` file. Boundary types are those of the archived `0/` dictionaries (Table 1).

test, while the forward-facing step exercises transient shock-capturing, conservation assessment, and controlled variations. The turbulent cube tests whether the assessment process can distinguish numerical completion from sufficiently developed flow. All three studies use OpenFOAM Foundation v14.

The assessment uses preserved simulation outputs and execution records. Case-specific criteria are applied to the underlying numerical evidence, while model-call and action records establish the role of agent reasoning. Reference agreement, numerical acceptance, and physical development are treated as distinct questions.

3.1 Converging-Diverging Nozzle

The canonical nozzle is an inviscid, compressible perfect-gas calculation using `shockFluid`. The gas properties are $\gamma = 1.4$ and $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$. The inlet stagnation pressure and temperature are $p_0 = 200$ kPa and $T_0 = 300$ K. The inlet, throat, and exit radii are respectively 0.0500, 0.0326, and 0.0354 m, with a total domain length of 0.33 m. The area ratio is $A_e/A^* = 1.179$ and the nominal

pressure ratio is $p_0/p_a = 6.67$.

A structured 5° wedge mesh contains 2,112 cells, with 132 axial and 16 radial divisions (Fig. 2a). Slip conditions are applied at the nozzle wall, and wedge boundaries represent axisymmetry. The inlet uses total-pressure and total-temperature conditions with an axially constrained velocity. The outlet uses zero-gradient conditions for pressure, temperature, and velocity. The nominal ambient pressure of 30 kPa is used in the operating-regime assessment; it is not imposed as a fixed outlet pressure, and the external jet is not simulated.

Quasi-one-dimensional theory supplies the initial internal fields and the analytical comparison. It does not constrain the subsequent CFD exit state. The numerical setup uses Kurganov fluxes [20], Minmod reconstruction, Euler time integration, and an adaptive timestep with a target maximum Courant number of 0.4. The canonical run advances to $t = 0.006$ s.

Assessment includes mesh validity, solver completion, positive and finite thermodynamic fields, conservation, stationarity, and consistency with the intended choked, supersonic-outlet regime. Registered requirements include a maximum steady mass-flow mismatch below 0.1% over the final 2 ms, normalized transient continuity error below 0.01%, and monitor drift and relative field changes below 0.2% over the same window. The outlet-pressure discrepancy from quasi-one-dimensional theory must remain below 5%, with the remaining registered theory-comparison quantities below 3%. These checks are combined with inlet-total-condition, sonic-throat ($0.9 < M_t < 1.1$), supersonic-outlet (minimum face-normal Mach number above 1.05), stagnation-enthalpy, and flux-consistency checks, twenty checks in total. The accepted parameterized run is classified as `PASS_SINGLE_MESH`; it is not itself a grid-convergence study.

3.2 Mach-2 Forward-Facing Step

The step demonstration solves the two-dimensional compressible Euler equations using `shockFluid`. The domain has length 3 and height 1, with the step beginning at $x = 0.6$ and rising to a height of 0.2. A structured mesh contains 16,128 hexahedral cells and one spanwise layer with `empty` front and back boundaries (Fig. 2b).

The normalized inlet and initial states are $p = 1$, $T = 1$, and $\mathbf{U} = (2, 0, 0)$; the normalized gas has unit sound speed, so the inlet Mach number is 2. Inlet values are fixed, the step surface is an inviscid slip boundary, and the upper and lower channel boundaries are symmetry planes. The outlet uses zero-gradient pressure with inlet-outlet treatment for velocity and temperature. The scheme uses Kurganov fluxes, van Leer reconstruction, and Euler time integration, copied unchanged from the Foundation forward-step tutorial recipe and hash-checked at run time. The target maximum Courant number is 0.2, and the canonical simulation advances to normalized time $t = 4$, with 40 saved states.

The assessment checks planar mesh topology, the prescribed initial state, realized supersonic inflow, field positivity, solver health, wall impermeability, and completion of the requested time horizon. Conservation is evaluated using the transient balance

$$\frac{d}{dt} \int_{\Omega} \rho dV + \int_{\partial\Omega} \rho \mathbf{U} \cdot \mathbf{n} dS = 0, \quad (1)$$

with discrete storage and numerical face-flux terms. The registered normalized closure residual must remain below 10^{-6} , with continuity across restart segments checked where applicable. Unequal instantaneous inlet and outlet fluxes are therefore not interpreted as a conservation failure when balanced by changing stored mass.

The compression diagnostic additionally requires measurable increases in pressure, density, and temperature across the detected front. These are directional physical checks, not a quantitative

reference-field comparison. The canonical Mach-2 result is assessed under this numerical and physical contract; an independent reference-field comparison is not part of its demonstrated acceptance. The case differs from the classical Mach-3 forward-facing-step problem [21] and is not presented as a reproduction of it.

3.3 Turbulent Cube Diagnostic Study

The cube study uses the Foundation-v14 `incompressibleFluid` solver with the unsteady Reynolds-averaged Navier–Stokes equations and k - ω SST closure [22]. With cube height $H = 1$, bulk velocity $U_b = 1$, and kinematic viscosity $\nu = 2.5 \times 10^{-5}$, the Reynolds number is $Re_H = U_b H / \nu = 40,000$, in a channel of height $2H$ comparable to the configuration of Martinuzzi and Tropea [23].

The computational domain spans $x/H \in [-6, 16]$, $y/H \in [0, 2]$, and $z/H \in [-12, 12]$. The cube occupies $x/H \in [0, 1]$, $y/H \in [0, 1]$, and $z/H \in [-0.5, 0.5]$. The mesh contains 567,360 hexahedral cells (Fig. 2c). Developed inlet velocity and turbulence profiles are mapped from a precursor channel calculation. The floor, roof, side walls, and cube use no-slip velocity conditions with the preserved SST wall-function treatment. The outlet uses fixed reference pressure and zero-gradient velocity and turbulence fields. Pressure is kinematic gauge pressure.

The calculation uses backward time integration, linear-upwind velocity convection, upwind turbulence convection, and a target maximum Courant number of 0.8. The preserved campaign continues a predecessor pilot and extends to $t^* = tU_b/H = 80$; force histories are analyzed from $t^* = 20$. Development is assessed using force histories, six wake probes, block-mean fields, and fluctuation measures rather than solver residuals alone.

The registered stationarity gate is applied to the pressure-plus-viscous force on the cube over the final 20 time units. It requires that the least-squares drift of each force component across the window remain below 2% of the mean streamwise force, that the mean lateral force remain below 5% of the streamwise force, and that the mean of $|F_z|$ over the second half of the window not exceed 1.25 times its first-half value. A supplementary, predeclared audit examines complete lateral-force cycles between successive upward zero crossings of F_z . For cycle n , the amplitude is

$$A_n = \frac{F_{z,\max,n} - F_{z,\min,n}}{2}, \quad (2)$$

and the audit criterion requires the relative change between successive comparable complete-cycle amplitudes to remain within 10%. Incomplete cycles are excluded from this comparison. Mean drag, probe statistics, and other development indicators provide complementary evidence. The cube is retained as a diagnostic stress test, not as an accepted turbulent reference solution.

3.4 Workflow Evaluation and Execution Provenance

The archived campaign includes nozzle geometry and operating-condition variations, together with step Mach-number, step-height, and execution-horizon variations. These records are examined for the sequence of diagnostic evidence, proposed action, permitted execution, and final assessment. They provide examples of acceptance, continuation, correction, and safe termination.

The nozzle and step runs record their model calls explicitly. Request interpretation, mesh-evidence review, diagnosis, and summary calls in these runs used `gemini-3.5-flash-lite`; the step runs additionally invoked a Gemini multimodal field-image observer whose model identifier was not recorded in the run provenance. The nozzle diagnosis step receives a theory-blind evidence packet from which analytical target values are excluded. These records are distinguished from deterministic-backend execution and replay of archived assessments. The cube campaign was executed outside the

agent loop and contains no model call; its analysis is attributed to quantitative diagnostic evidence, and the available trace does not establish that an LLM independently discovered the growing wake mode.

The present evaluation documents demonstrated workflow behavior and scientific decisions. Archived-case replay is used to check consistency of recorded outcomes, not as evidence of new autonomous CFD execution or generalization. Controlled ablations against fixed-rule workflows have not yet been performed; consequently, the study does not claim a measured improvement in success rate, computational cost, or accuracy attributable to the LLM.

4 Results

4.1 Converging–Diverging Nozzle

The canonical nozzle run completed to $t = 6$ ms in 30,471 adaptive steps (maximum Courant number 0.400) and passed all twenty registered checks, yielding `PASS_SINGLE_MESH` and a deterministic `ACCEPT`. Figure 3 compares radially volume-weighted axial profiles with the isentropic quasi-one-dimensional solution for the same area distribution. The flow accelerates from $M = 0.253$ in the inlet section, reaches $M = 1.029$ in the throat, and leaves the nozzle at $M = 1.513$, while static pressure and temperature fall to $0.266 p_0$ and $0.686 T_0$. The minimum slab-averaged Mach number downstream of the throat is 1.17 and the minimum face-normal outlet Mach number is 1.46, so no internal normal shock forms and the entire exit plane is supersonic. The exit static pressure, 53.1 kPa, exceeds the 30 kPa nominal ambient pressure, which is consistent with an underexpanded nozzle.

Table 2 lists the deterministic comparison with theory. All differences are below 2%; the largest is the exit static pressure (−1.83%, limit 5%), and the mass flow is 1.11% below the ideal choked value. These differences are consistent with a two-dimensional axisymmetric solution that departs from strictly one-dimensional flow near the conical corners. The inlet total conditions are reproduced to a relative error of 2.5×10^{-8} in p_0 and 7.2×10^{-9} in T_0 . Over the final 2 ms, the maximum steady mass-flow mismatch is 0.055%, the largest monitor drift is 0.081%, the largest relative field change is 0.111%, and the maximum transient continuity error is $6.4 \times 10^{-8}\%$.

A separate archived run of the same specification illustrates the corrective loop (Fig. 3c). Executed initially only to $t = 1$ ms, it passed 17 of 20 checks but failed the steady mass-balance (1.35% against 0.1%), monitor-stationarity (2.0% against 0.2%), and field-stationarity (1.2% against 0.2%) criteria. At this endpoint the exit-state differences from quasi-one-dimensional theory were already below 0.7%. Agreement with the reference therefore did not establish that the flow had settled, and the stationarity evidence determined the outcome. After an approved continuation to 6 ms, all three metrics fell below their limits (ratios 0.59, 0.41, and 0.66) and the run was accepted.

4.2 Mach-2 Forward-Facing Step

The canonical step run completed to $t = 4$ in 5,428 steps and 102 s of wall time, passing all 17 hard checks and the three compression checks (`PASS_2D_FORWARD_STEP`, deterministic `ACCEPT`). The realized inlet Mach number was 2.000006, and the maximum Courant number was 0.202 against the 0.2 controller target. Figure 4 shows the flow at the registered horizon. A detached compression front spans the channel ahead of the step, an expansion fan forms at the step shoulder, and an oblique shock from the shoulder region reflects from the upper symmetry plane near $x \approx 1.95$. Across the detected front, pressure, density, and temperature rise by factors of 5.18, 2.87, and 1.81. These values exceed the stationary normal-shock ratios for Mach 2 (4.50, 2.67, and 1.69); the

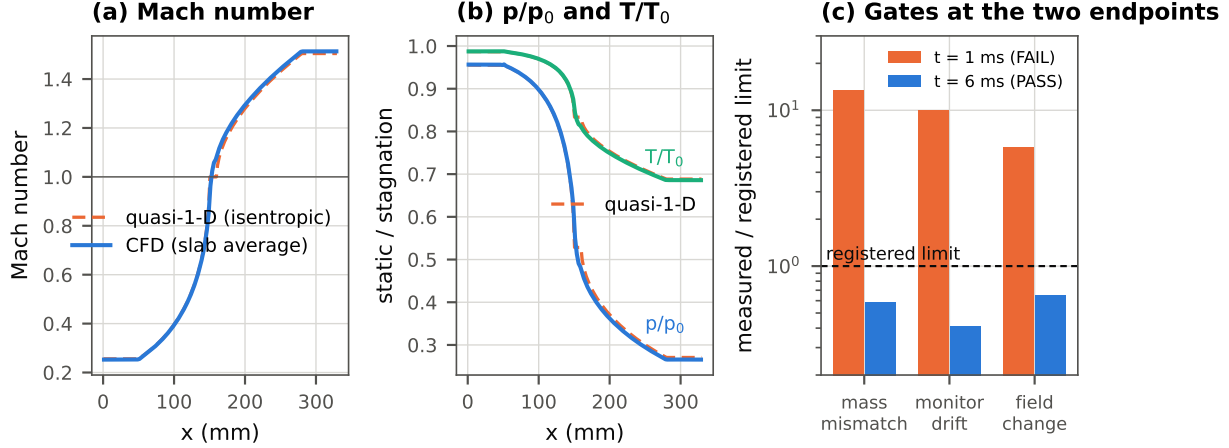


Figure 3: Nozzle results. (a,b) Axial profiles of the canonical run at $t = 6$ ms (solid) and the isentropic quasi-one-dimensional solution for the registered area distribution (dashed). (c) The three stationarity checks of the separate continuation run, expressed as measured value over registered limit, at the initial 1 ms endpoint (failed) and after the approved continuation to 6 ms (passed).

difference is qualitatively consistent with a front that is moving upstream, but these gradient-based regional measurements are not grid-converged and are not used as acceptance bands.

The detected front moves upstream monotonically, from $x = 0.525$ at $t = 0.1$ to $x = 0.113$ at $t = 4$ (Fig. 4c). A quasi-one-dimensional estimate explains this behavior. Behind a normal shock at Mach 2 the flow has $M = 0.577$, whose sonic area is 0.822 of the channel height. This exceeds the 0.8 opening above the step, so the subsonic post-shock flow cannot pass the contraction and the front is displaced upstream. At Mach 3 the corresponding sonic area is 0.719, which is why a standing detached shock is possible in the classical Mach-3 configuration. The canonical Mach-2 solution is therefore genuinely transient at $t = 4$. The registered contract evaluates health, conservation, and compression structure at that horizon and applies no stationarity criterion to this family.

Mass conservation closes to round-off in the storage-aware form of Eq. (1). The maximum normalized closure residual is 5.7×10^{-11} (99th percentile 3.0×10^{-11}), against the registered 10^{-6} limit (Fig. 4d). The cumulative defect is 8×10^{-14} of the initial mass, and the largest flux through the impermeable boundaries is 1.1×10^{-17} . At $t = 4$ the instantaneous outlet mass flux is 16% below the inlet flux, a difference balanced by mass accumulating behind the advancing front. A test based on inlet-outlet equality would have rejected a correctly conservative transient solution.

4.3 Turbulent Cube Diagnostic Study

The cube calculation completed its bounded extension to $t^* = 80$ without numerical failure. In the final two segments the maximum Courant number was 0.79, the largest reported continuity error was 2×10^{-12} , and the outlet and inlet volume fluxes balanced to a relative 3×10^{-11} . Instantaneous fields at $t^* = 80$ show a physically plausible horseshoe region, separated shear layers from the leading edges, and a recirculating wake (Fig. 5a). The streamwise force also appears settled. Over the registered window $t^* \in [59.98, 79.98]$, the mean drag coefficient is $C_D = 2F_x = 1.412$ and its drift is 0.075% of the mean, well within the 2% limit, and the mean lateral force is 0.063% of the drag (limit 5%).

The lateral force shows that the flow has not developed (Fig. 5c). F_z oscillates with an almost

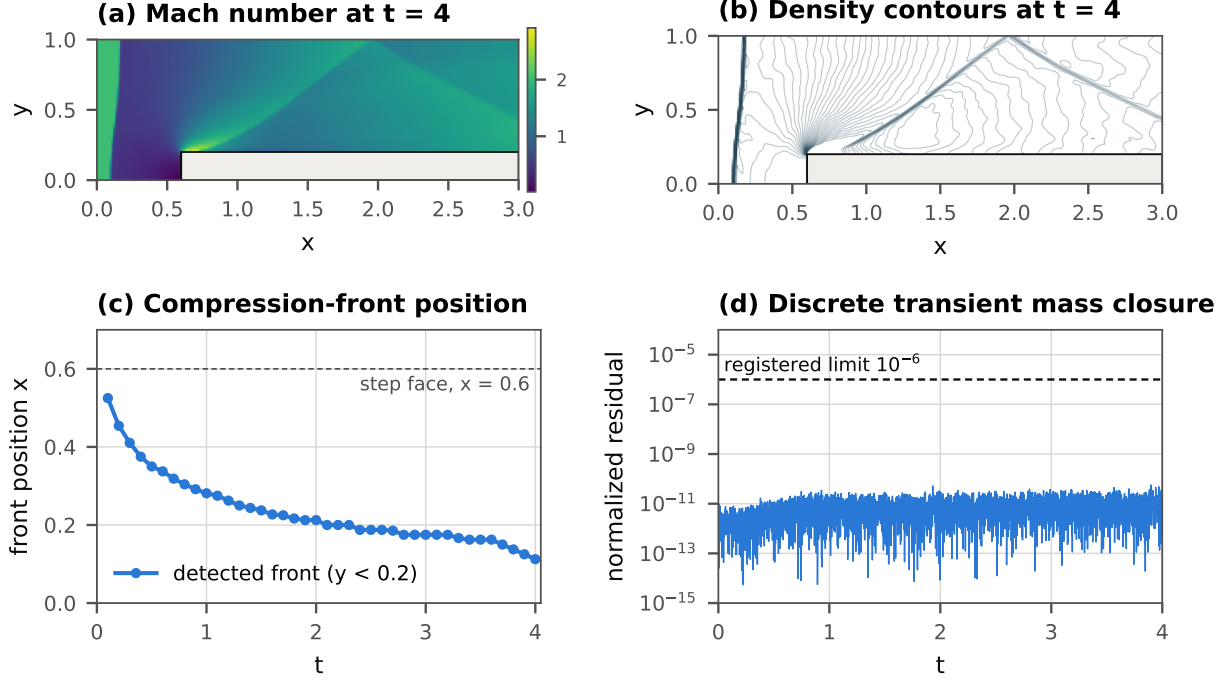


Figure 4: Mach-2 forward-facing step. (a) Mach number and (b) density contours at $t = 4$, reproduced from the archived pipeline renderings of the native fields with new axes. (c) Position of the detected compression front below the step height. (d) Normalized discrete closure residual of Eq. (1) at every time step, with the registered limit. **TODO:** re-render (a,b) directly from the native $t = 4$ fields once they are exported from the WSL case directory.

constant period of about 10.1 time units ($St \approx 0.098$), while its envelope grows: the mean of $|F_z|$ rises from 1.83×10^{-3} in the first half of the window to 3.85×10^{-3} in the second, a ratio of 2.10 against the registered limit of 1.25. The gate therefore returns `STILL_DEVELOPING`, and the run is rejected. The supplementary complete-cycle audit agrees (Fig. 5d). Successive complete-cycle amplitudes grow by 209%, 137%, and 113%, and the last comparable pair exceeds the 10% audit limit by an order of magnitude. The growth rate is declining but the mode has not saturated. The two diagnostics are different measurements and are reported separately. Only the half-window ratio is part of the registered gate.

The result illustrates why the assessment cannot rely on a single signal. Solver completion, small continuity errors, a stable mean drag, and plausible instantaneous contours all coexist here with a lateral mode whose amplitude more than doubles every cycle. Block-averaged fields are also misleading: between the $t^* = 60$ –70 and 70–80 averages the mean drag changes by only 0.04% and wake-probe means by at most $0.002 U_b$, while the root-mean-square lateral force increases by 110%. Period averages suppress an alternating signal whose amplitude is still growing.

4.4 Agentic Workflow and Corrective Actions

Table 3 summarizes the registered outcomes, and Table 4 lists recorded decision points in which a model proposal met the deterministic action validator. In each case the proposed action was consistent with the deterministic evidence and was approved. In the nozzle continuation, the model read the stationarity failures and proposed `CONTINUE_RUN`. In a Mach-3 step run that was healthy

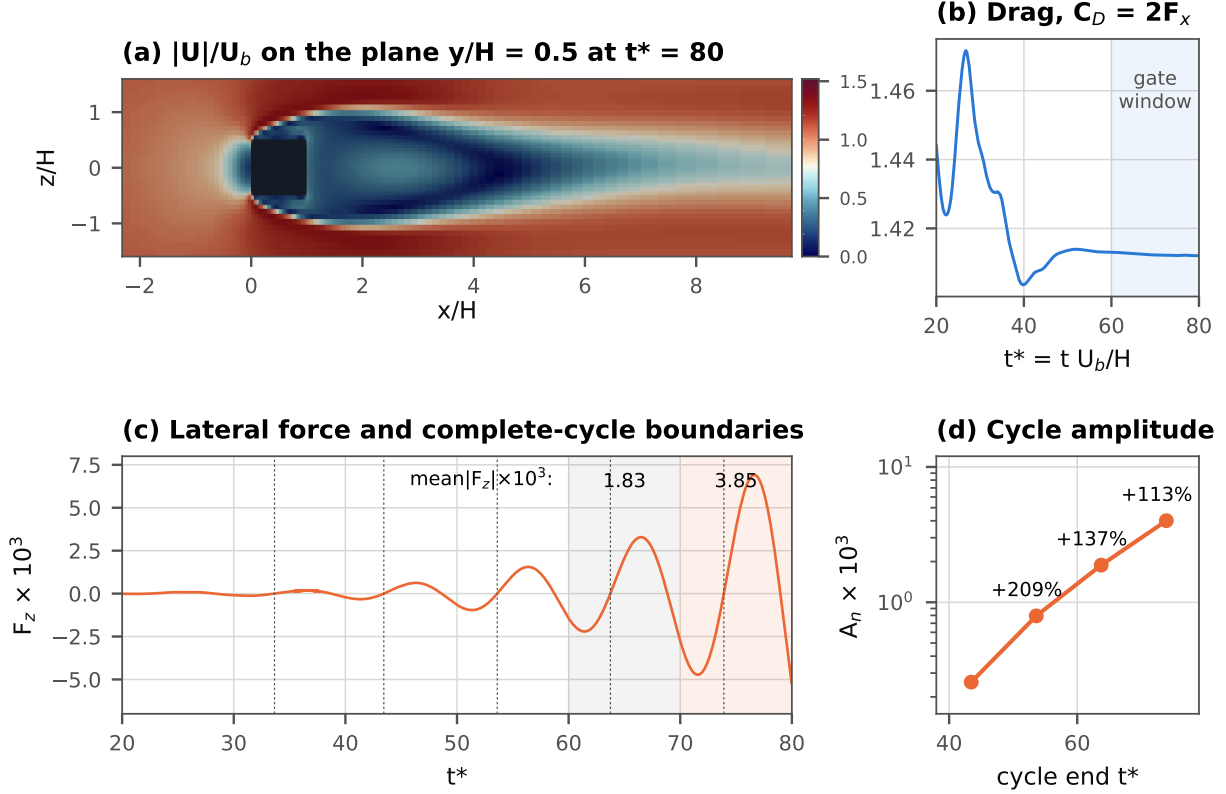


Figure 5: Turbulent cube diagnostic study. (a) Velocity magnitude on the plane $y/H = 0.5$ at $t^* = 80$, from the archived ParaView rendering of the native field with new axes. (b) Drag coefficient history; the shaded band is the registered 20-unit window. (c) Lateral force, with the two half-windows of the registered growth test shaded and the mean $|F_z|$ of each labelled; dotted lines mark the upward zero crossings that bound complete cycles. (d) Complete-cycle amplitudes from Eq. (2) and their successive changes.

but had reached only $t = 2$ of the registered $t = 4$ horizon, the model proposed extending the end time, which the validator permitted as a bounded change ($2 \rightarrow 4$) before reassessment and acceptance. In a Mach-3 run with a taller step ($h = 0.3$), the deterministic compression diagnostic could no longer measure a front; the model attributed this to upstream migration of the shock system and proposed **FAIL_SAFELY**. This outcome is consistent with the choking estimate above, since the post-shock sonic area at Mach 3 (0.719) exceeds the 0.7 opening. That run was preserved and rejected rather than tuned.

These records demonstrate that the interface between model reasoning and deterministic authority functions as designed. In every recorded case the terminal verdict was computed by the validators, and continuation or extension occurred only through permitted actions. The records do not measure whether model reasoning improves on a fixed rule. In these examples a rule keyed to the failed check would have selected the same action, and the multimodal observer returned no usable observation in some runs without affecting the outcome.

5 Limitations

The demonstration covers three CFD studies. Nozzle and forward-facing-step flows are the validated, routable families. The cube is supporting diagnostic evidence and is not routable. Each new physics family requires its own scientific contract, and acceptance under a family contract is not a general statement of CFD correctness. The accepted runs are single-mesh verifications. Grid and time-step independence are not re-established per case, and the step case has no quantitative reference-field comparison. The cube wall treatment is uncertain: at $t^* = 80$ the cube-surface y^+ has a median of 24.7 and a range of 2.8–96.5, with 67% of the area below $y^+ = 30$. A provisional comparison of its evolving mean wake with ERCOFTAC centre-line profiles is not satisfactory, so no experimental validation is claimed. General CAD/STEP-to-CFD execution is not demonstrated. Finally, the evaluation documents workflow behavior rather than an ablation. Comparisons with a fixed-rule controller and with model-only acceptance remain future work.

6 Conclusion

CFD Forge links engineering requests to OpenFOAM case construction, execution, diagnosis, correction, and reporting, while keeping scientific acceptance in deterministic, family-specific contracts. Language-model reasoning interprets requests and evidence and proposes bounded actions; validators decide whether those actions are admissible and whether a result is accepted. Under this division, a converging–diverging nozzle was accepted with sub-2% agreement with quasi-one-dimensional theory after its stationarity evidence, not its reference agreement, determined when the run was complete. A transient Mach-2 forward-facing-step flow was accepted with storage-aware mass closure at round-off. A three-dimensional turbulent cube was rejected because a growing lateral mode showed that solver completion and a settled mean drag did not establish a developed flow. Case files, model calls, action rulings, and field-evolution records keep each decision auditable. These elements provide a basis for high-throughput simulation, active learning, and design-optimization loops in which automated evaluations must remain scientifically accountable.

References

- [1] J. P. Slotnick, A. Khodadoust, J. Alonso, D. Darmofal, W. Gropp, E. Lurie, and D. J. Mavriplis, *CFD Vision 2030 Study: A Path to Revolutionary Computational Aerosciences*. NASA, Technical Report NASA/CR-2014-218178, 2014. <https://ntrs.nasa.gov/citations/20140003093>.
- [2] H. G. Weller, G. Tabor, H. Jasak, and C. Fureby, “A tensorial approach to computational continuum mechanics using object-oriented techniques,” *Computers in Physics*, vol. 12, no. 6, pp. 620–631, 1998. DOI: <https://doi.org/10.1063/1.168744>.
- [3] A. I. J. Forrester and A. J. Keane, “Recent advances in surrogate-based optimization,” *Progress in Aerospace Sciences*, vol. 45, nos. 1–3, pp. 50–79, 2009. DOI: <https://doi.org/10.1016/j.paerosci.2008.11.001>.
- [4] Y. Chen, L. Zhang, X. Zhu, H. Zhou, and Z. Ren, “Large language model-assisted sensitivity analysis and parameter optimization in computational fluid dynamics,” *Theoretical and Applied Mechanics Letters*, Article 100660, 2026, advance online publication. DOI: <https://doi.org/10.1016/j.taml.2026.100660>.

- [5] S. L. Brunton, B. R. Noack, and P. Koumoutsakos, “Machine learning for fluid mechanics,” *Annual Review of Fluid Mechanics*, vol. 52, pp. 477–508, 2020. DOI: <https://doi.org/10.1146/annurev-fluid-010719-060214>.
- [6] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, “ReAct: Synergizing reasoning and acting in language models,” in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. Available at <https://arxiv.org/abs/2210.03629>.
- [7] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, and T. Scialom, “Toolformer: Language models can teach themselves to use tools,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023. DOI: <https://doi.org/10.52202/075280-2997>.
- [8] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020. Available at <https://arxiv.org/abs/2005.11401>.
- [9] Y. Chen, X. Zhu, H. Zhou, and Z. Ren, “MetaOpenFOAM: An LLM-based multi-agent framework for CFD,” *arXiv preprint arXiv:2407.21320*, 2024. DOI: <https://doi.org/10.48550/arXiv.2407.21320>.
- [10] Y. Chen, X. Zhu, H. Zhou, and Z. Ren, “MetaOpenFOAM 2.0: Large language model driven chain of thought for automating CFD simulation and post-processing,” *arXiv preprint arXiv:2502.00498*, 2025. DOI: <https://doi.org/10.48550/arXiv.2502.00498>.
- [11] S. Pandey, R. Xu, W. Wang, and X. Chu, “OpenFOAMGPT: A retrieval-augmented large language model (LLM) agent for OpenFOAM-based computational fluid dynamics,” *Physics of Fluids*, vol. 37, no. 3, Article 035120, 2025. DOI: <https://doi.org/10.1063/5.0257555>.
- [12] J. Feng, R. Xu, and X. Chu, “OpenFOAMGPT 2.0: End-to-end, trustworthy automation for computational fluid dynamics,” *International Journal of Heat and Fluid Flow*, vol. 120, Article 110399, 2026. DOI: <https://doi.org/10.1016/j.ijheatfluidflow.2026.110399>.
- [13] Z. Dong, Z. Lu, and Y. Yang, “Fine-tuning a large language model for automating computational fluid dynamics simulations,” *Theoretical and Applied Mechanics Letters*, vol. 15, Article 100594, 2025. DOI: <https://doi.org/10.1016/j.taml.2025.100594>.
- [14] L. Yue, N. Somasekharan, T. Zhang, Y. Cao, Z. Chen, S. Di, and S. Pan, “Foam-Agent: A large language model-based multi-agent framework for automating computational fluid dynamics workflows,” *Computer Methods in Applied Mechanics and Engineering*, vol. 461, Article 119271, 2026. DOI: <https://doi.org/10.1016/j.cma.2026.119271>.
- [15] E. Fan, K. Hu, Z. Wu, J. Ge, J. Miao, Y. Zhang, H. Sun, W. Wang, and T. Zhang, “ChatCFD: A large language model-driven agent for end-to-end computational fluid dynamics automation with structured knowledge and reasoning,” *Advanced Intelligent Discovery*, vol. 2, no. 3, Article e202500174, 2026. DOI: <https://doi.org/10.1002/aidi.202500174>.
- [16] W. L. Oberkampf and T. G. Trucano, “Verification and validation in computational fluid dynamics,” *Progress in Aerospace Sciences*, vol. 38, no. 3, pp. 209–272, 2002. DOI: [https://doi.org/10.1016/S0376-0421\(02\)00005-2](https://doi.org/10.1016/S0376-0421(02)00005-2).

- [17] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad, “Procedure for estimation and reporting of uncertainty due to discretization in CFD applications,” *Journal of Fluids Engineering*, vol. 130, no. 7, Article 078001, 2008. DOI: <https://doi.org/10.1115/1.2960953>.
- [18] N. Somasekharan, R. Pathak, M. Dhanakoti, T. Zhang, L. Yue, A. Zhu, and S. Pan, “AI CFD Scientist: Toward open-ended computational fluid dynamics discovery with physics-aware AI agents,” *arXiv preprint arXiv:2605.06607*, 2026. DOI: <https://doi.org/10.48550/arXiv.2605.06607>.
- [19] E. Fan, L. Shi, Z. Li, and C.-y. Wen, “PhyNiKCE: A neurosymbolic agentic framework for autonomous computational fluid dynamics,” *arXiv preprint arXiv:2602.11666*, 2026. DOI: <https://doi.org/10.48550/arXiv.2602.11666>.
- [20] A. Kurganov and E. Tadmor, “New high-resolution central schemes for nonlinear conservation laws and convection–diffusion equations,” *Journal of Computational Physics*, vol. 160, no. 1, pp. 241–282, 2000. DOI: <https://doi.org/10.1006/jcph.2000.6459>.
- [21] P. Woodward and P. Colella, “The numerical simulation of two-dimensional fluid flow with strong shocks,” *Journal of Computational Physics*, vol. 54, no. 1, pp. 115–173, 1984. DOI: [https://doi.org/10.1016/0021-9991\(84\)90142-6](https://doi.org/10.1016/0021-9991(84)90142-6).
- [22] F. R. Menter, “Two-equation eddy-viscosity turbulence models for engineering applications,” *AIAA Journal*, vol. 32, no. 8, pp. 1598–1605, 1994. DOI: <https://doi.org/10.2514/3.12149>.
- [23] R. Martinuzzi and C. Tropea, “The flow around surface-mounted, prismatic obstacles placed in a fully developed channel flow,” *Journal of Fluids Engineering*, vol. 115, no. 1, pp. 85–92, 1993. DOI: <https://doi.org/10.1115/1.2910118>.

Table 1: Simulation setup. All cases use OpenFOAM Foundation v14 through `foamRun`. The scheme, solution, thermophysical and mesh files quoted here match the SHA-256 digests recorded for the native cases in the field-evolution manifests; cube boundary types are taken from the preserved initial-condition audit of its predecessor pilot.

	Nozzle	Forward-facing step	Surface-mounted cube
Model	Inviscid compressible Euler, perfect gas ($\gamma = 1.4$, $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$); <code>shockFluid</code>	Inviscid compressible Euler, normalized perfect gas ($\gamma = 1.4$, sound speed 1); <code>shockFluid</code>	Incompressible URANS, k - ω SST [22], $\nu = 2.5 \times 10^{-5}$, $Re_H = 40,000$; <code>incompressibleFluid</code>
Mesh	5° wedge, $132 \times 16 = 2,112$ hexahedra; max. aspect ratio 2.46, non-orthogonality 9.6° , skewness 0.33	Three blocks, uniform $\Delta x = \Delta y = 0.0125$, 16,128 hexahedra, one <code>empty</code> spanwise layer	Rectilinear, $116 \times 40 \times 124$ minus the 20^3 cube cells = 567,360 hexahedra; max. aspect ratio 18.4, six partitions
Inlet	<code>totalPressure</code> $p_0 = 200 \text{ kPa}$, <code>totalTemperature</code> $T_0 = 300 \text{ K}$, axial-direction velocity	Fixed $p = 1$, $T = 1$, $\mathbf{U} = (2, 0, 0)$	Developed $\mathbf{U}$, k , ω mapped from a precursor channel; $\partial p / \partial n = 0$
Outlet	Zero gradient for p , T , $\mathbf{U}$ (30 kPa ambient not imposed)	Zero-gradient p ; <code>inletOutlet</code> $\mathbf{U}$, T	Fixed gauge p ; zero gradient for $\mathbf{U}$, k , ω
Other boundaries	Slip wall; <code>wedge</code> front/back; axis	Step: slip; top/bottom: <code>symmetryPlane</code>	Floor, roof ($y/H = 2$), side walls ($z/H = \pm 12$), cube: no-slip with <code>kqR/omega/nutUSpalding</code> wall functions
Numerics	Kurganov flux [20], Minmod reconstruction, Euler time step, $Co_{\max} = 0.4$	Kurganov flux, van Leer (ρ , T) and van Leer-V ($\mathbf{U}$) reconstruction, Euler, $Co_{\max} = 0.2$	Backward time step, PIMPLE (2 outer, 2 pressure correctors), linear-upwind $\mathbf{U}$, upwind k, ω , $Co_{\max} = 0.8$, $\Delta t \leq 0.02$
Horizon	$t = 6 \text{ ms}$ (30,471 steps)	$t = 4$ (5,428 steps)	$t^* = 80$
Assessment basis	Quasi-1-D analytical consistency and registered numerical checks	Registered numerical checks and directional compression diagnostics	Registered stationarity/development gate

Table 2: Canonical nozzle: CFD versus isentropic quasi-one-dimensional theory at $t = 6 \text{ ms}$. Exit values are area averages on the outlet patch; throat and inlet values are slab averages. Mass flow is scaled from the wedge to the full nozzle.

Quantity	CFD	Quasi-1-D	Difference	Registered limit
Inlet Mach number ^a	0.2528	0.2558	-1.2%	-
Throat Mach number	1.029	1	-	$0.9 < M_t < 1.1$
Exit Mach number	1.5133	1.5044	+0.59%	< 3%
Exit static pressure (kPa)	53.14	54.13	-1.83%	< 5%
Exit static temperature (K)	205.76	206.52	-0.37%	< 3%
Exit velocity (m s^{-1})	435.09	433.36	+0.40%	< 3%
Mass flow (kg s^{-1})	1.541	1.558	-1.11%	< 3%

^aNot a registered check; theory value computed from the inlet area ratio.

Table 3: Registered scientific assessment of the three studies.

Evidence	Nozzle	Forward-facing step	Cube
Mesh	checkMesh OK	checkMesh OK; 2 solution directions	checkMesh OK
Solver health	Completed; $p, T, \rho > 0$ at all saved times	Completed; positive at every step; no fatal signature	Completed; no warnings in final segments
Conservation	Window mismatch 0.055% ($< 0.1\%$)	Closure residual 5.7×10^{-11} ($< 10^{-6}$)	Volume-flux imbalance 3×10^{-11}
Physical or reference	Quasi-1-D differences $\leq 1.83\%$	p, ρ, T rise across front	Not assessed (no validated reference)
Stationarity or development	Drift 0.081%, field change 0.111% ($< 0.2\%$)	Not required (transient family)	Lateral growth ratio 2.10 (> 1.25)
Registered outcome	ACCEPT (PASS_SINGLE_MESH)	ACCEPT (PASS_2D_FORWARD_STEP)	REJECT (STILL_DEVELOPING)

Table 4: Recorded decision points with genuine model calls (`gemini-3.5-flash-lite`). “Approved” denotes a ruling of the deterministic action validator; the verdict column is computed by the scientific validator.

Run and state	Deterministic evidence	Model proposal	Validator	Verdict
Nozzle, $t = 1$ ms	17/20 checks; mass mismatch 1.35%, drift 2.0%, field change 1.2%	CONTINUE_RUN	Approved	FAIL $\rightarrow$ continue
Nozzle, $t = 6$ ms	20/20 checks	ACCEPT	Approved	PASS_SINGLE_MESH
Step, Mach 2, $t = 4$	17/17 hard checks; compression rises measured	ACCEPT	Approved	PASS_2D_FORWARD_STEP
Step, Mach 3, $t = 2$ of 4	Hard checks pass; registered horizon not reached	EXTEND_END_TIME (2 $\rightarrow$ 4)	Approved	Healthy, not final $\rightarrow$ extend
Step, Mach 3, $t = 4$	17/17 hard checks	ACCEPT	Approved	PASS_2D_FORWARD_STEP
Step, Mach 3, $h = 0.3$	Compression front not measurable	FAIL_SAFELY	Approved	FAIL (preserved)